Differentiation Rules: Power, Product, Quotient, Chain

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For parametric $x = e^t \cos t$, $y = e^t \sin t$, find $\frac{d^2y}{dx^2}$ at $t = 0$.

First, $\frac{dy}{dx} = \frac{\sin t + \cos t}{\cos t - \sin t}$. Then $\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{2}{(\cos t - \sin t)^2}$. Dividing by $\frac{dx}{dt} = e^t(\cos t - \sin t)$ at $t = 0$ gives $\frac{d^2y}{dx^2}\big|_{t=0} = 2$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $\sin(\theta) = -\frac{24}{25}$ and θ lies in quadrant III, find $\sin 2\theta$.

In quadrant III, $\cos(\theta) = -\frac{7}{25}$ (from $\sin^2\theta + \cos^2\theta = 1$ with the quadrant's sign). Then $\sin 2\theta = \frac{336}{625}$ using the double-angle formula.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $\sin(\theta) = \frac{5}{13}$ and θ lies in quadrant II, find $\cos 2\theta$.

In quadrant II, $\cos(\theta) = -\frac{12}{13}$ (from $\sin^2\theta + \cos^2\theta = 1$ with the quadrant's sign). Then $\cos 2\theta = \frac{119}{169}$ using the double-angle formula.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Coffee at 70°C is placed in a 20°C room and cools to 60°C in 10 min (Newton's law of cooling). How long (min) until it reaches 40°C?

$T - 20 = 50e^{-kt}$ with $k = \frac{\ln\left(\frac{5}{4}\right)}{10}$. Solve $50e^{-kt} = 20$: $t = \frac{10\ln\left(\frac{5}{2}\right)}{\ln\left(\frac{5}{4}\right)}$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Water pours into an inverted cone (radius 3 m, height 9 m) at 3 m³/min. How fast is the depth rising when it is 8 m? (m/min)

By similar triangles $r = \frac{h}{3}$, so $V = \frac{\pi h^3}{27}$. Then $\frac{dV}{dt} = \pi\frac{9}{81}h^2\frac{dh}{dt}$, which gives $\frac{27}{64\pi}$ m/min at $h = 8$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Product rule for $f = x^2e^x$:.

Final Answer: $e^x(x^2 + 2x)$. Applying Derivatives principles, $2xe^x + x^2e^x = e^x(x^2 + 2x)$. Therefore, the verified correct option is $e^x(x^2 + 2x)$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

$\frac{d}{dx}[f(g(x))]$ equals (chain rule):.

Final Answer: $f'(g(x)) \cdot g'(x)$. Chain rule: outer derivative evaluated at inner × inner derivative.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Quotient Rule: $\frac{d}{dx}\left(\frac{x}{x+1}\right) =$.

Final Answer: $\frac{1}{(x+1)^2}$. $\frac{1 \cdot (x+1) - x \cdot 1}{(x+1)^2} = \frac{1}{(x+1)^2}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Using the Product Rule, $\frac{d}{dx}[(3x+1)(x^2-2)] =$.

Final Answer: all are equivalent. All three expressions are equivalent: $9x^2 + 2x - 6$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Product of two power series $A(x) = \sum a_n x^n$ and $B(x) = \sum b_n x^n$: coefficient of x^n in $A(x)B(x)$ is:.

Final Answer: $\sum_{k=0}^{n} a_k b_{n-k}$ (Cauchy product). Applying Sequences and Series principles, Cauchy product convolution. Therefore, the verified correct option is $\sum_{k=0}^{n} a_k b_{n-k}$ (Cauchy product).

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 1 · FoundationsOn an AP Calculus BC exam question, a student writes $\int_0^5 (R(t) - D(t))\,dt = 42.18$ as their final response to a question asking for the total amount of water in a tank at $t = 5$, given initial volume 100 gallons. Identify the missing component in the student's solution.

The student computed only the net change $\int_0^5 (R(t) - D(t))\,dt$, failing to add the initial condition $W(0) = 100$. By the Fundamental Theorem of Calculus: $W(5) = W(0) + \int_0^5 (R(t) - D(t))\,dt = 100 + 42.18 =$ **142.18** gallons. Failing to include the initial condition is a common fatal error on FRQ Question 1.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · Advanced / AP FRQLet $f(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$. Identify the closed-form elementary function that this series represents, and determine its radius of convergence.

Write out the terms: $f(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \ldots$ Recall the definition of the hyperbolic sine function: $\sinh(x) = \frac{e^x - e^{-x}}{2}$. Expanding e^x and e^{-x} shows that only odd powers remain with coefficients $1/(2n+1)!$. Thus $f(x) = \sinh$**(x)**. Apply the Ratio Test: \[\lim_{n\to\infty} \frac{x^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{x^{2n+1}} = x^2 \lim_{n\to\infty} \frac{1}{(2n+3)(2n+2)} = 0

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · AdvancedA differential equation $\frac{dy}{dx} = (y-1)^2\cos(x)$ has initial condition $y(0) = 2$. Find the particular solution $y = f(x)$.

Separate variables: $\frac{dy}{(y-1)^2} = \cos(x)\,dx$. Integrate both sides: $-\frac{1}{y-1} = \sin(x) + C$. Apply initial condition $x = 0, y = 2$: $-\frac{1}{2-1} = \sin(0) + C \Rightarrow -1 = C$. Substitute $C = -1$:

$-\frac{1}{y-1} = \sin(x) - 1 \Rightarrow \frac{1}{y-1} = 1 - \sin(x) \Rightarrow y - 1 = \frac{1}{1-\sin(x)} \Rightarrow \mathbf{y = 1 + \frac{1}{1-\sin(x)}}$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 2 · MediumThe graph of f' consists of a semicircle of radius 2 centered at the origin, lying above the x-axis. If $f(0) = 5$, find the value of $f(2)$.

By the Fundamental Theorem of Calculus: $f(2) = f(0) + \int_0^2 f'(x)\,dx$. The region under f' from $x = 0$ to $x = 2$ is a quarter-circle of radius 2: $\int_0^2 f'(x)\,dx = \frac{1}{4}\pi(2^2) = \pi$. Therefore, $f(2) = 5 + \pi \approx \mathbf{8.142}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 12.01 Convergence Tests

Level 1 · FoundationsOn a Rate In / Rate Out FRQ, if rain falls into a cistern at rate $R(t)$ and is pumped out at rate $P(t)$, when is the amount of water in the cistern at a relative maximum?

Let $W(t)$ be the amount of water. $W'(t) = R(t) - P(t)$. A relative maximum occurs when $W'(t) = 0$ and changes from positive to negative, which occurs when $R(t) = P(t)$ and $R(t)$ crosses below $P(t)$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 12.02 Power Series and Taylor Series

Level 1 · FoundationsHow many total questions are on the AP Calculus AB exam, and what percentage of the composite score comes from Free-Response questions?

The exam consists of 45 Multiple-Choice Questions and 6 Free-Response Questions (51 total). Free Response accounts for exactly 50% of the composite score (54 out of 108 scaled points).

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

[AP Calculus AB Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

[AP Calculus BC Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 15.01 Double Integrals

Level 2 · Medium Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.

Step 1: Test for absolute convergence. Consider $\sum |a_n| = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$. This is a p-series with $p = 1/2 \le 1$, which diverges. Hence the series does NOT converge absolutely. Step 2: Apply Alternating Series Test. Here $b_n = \frac{1}{\sqrt{n}} > 0$. 1. Monotonicity: \(n+1 > n \implies \sqrt{n+1} > \sqrt{n} \implies \frac{1}{\sqrt{n+1}} conditionally convergent.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Level 1 · Foundations Evaluate $\iint_R (2x + y)\,dA$ where R is the rectangular region $[0, 3] \times [1, 2]$.

Set up iterated integral: $\int_1^2 \int_0^3 (2x + y)\,dx\,dy = \int_1^2 [x^2 + xy]_0^3 dy = \int_1^2 (9 + 3y)\,dy$ Integrate with respect to y: $\left[9y + \frac{3y^2}{2}\right]_1^2 = (18 + \frac{12}{2}) - (9 + \frac{3}{2}) = 24 - 10.5 = 13.5 = \frac{27}{2}$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the derivative of $f(x) = 4x^5 - 3x^3 + 7x - 9$.

a. $20x^4 - 9x^2 + 7$ ✓

b. $20x^4 - 9x^2$

c. $4x^4 - 3x^2 + 7$

d. $20x^4 - 6x^2 + 7$

By the power rule: $\frac{d}{dx}(4x^5) = 20x^4$, $\frac{d}{dx}(-3x^3) = -9x^2$, $\frac{d}{dx}(7x) = 7$, $\frac{d}{dx}(-9) = 0$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find all points where the curve $y = x^3 - 3x^2 - 9x + 5$ has a horizontal tangent.

a. $x = -1$ **and** $x = 3$ ✓

b. $x = 1$ and $x = -3$

c. $x = 0$ and $x = 3$

d. $x = -1$ only

$y' = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1) = 0 \Rightarrow x = 3$ or $x = -1$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the second derivative y'' for $y = \cos(2x)$.

a. $-4\cos(2x)$ ✓

b. $4\cos(2x)$

c. $-2\sin(2x)$

d. $-4\sin(2x)$

$y' = -2\sin(2x)$. $y'' = \frac{d}{dx}[-2\sin(2x)] = -2(2\cos(2x)) = -4\cos(2x)$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Implicit differentiation: $\frac{d^2y}{dx^2}$ at $(1, 1)$ for $x^2 + y^2 = 2$:

a. -1

b. -2 ✓

c. 0

d. 1

Differentiating $x^2 + y^2 = 2$ gives $y' = -\frac{x}{y}$ and $y'' = -\frac{y - xy'}{y^2} = -\frac{x^2 + y^2}{y^3}$. At $(1, 1)$, $y'' = -\frac{2}{1^3} = -2$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Cauchy product: $(\sum_{n=0}^{\infty} x^n)^2 = \sum_{n=0}^{\infty} (n + 1)x^n$. This equals:

a. $1/(1 - x)^2$ ✓

b. $1/(1 - x)$

c. $(1 - x)^2$

d. $2/(1 - x)$

Applying Convergence Tests and Power Series principles, $(1/(1 - x))^2 = \sum(n + 1)x^n$. Therefore, the verified correct option is $1/(1 - x)^2$.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Implicit differentiation of $F(x,y)=0$: $\frac{dy}{dx}=$

a. $-F_x/F_y$ ✓

b. F_x/F_y

c. $-F_y/F_x$

d. F_{xy}

Applying Applications of Derivatives principles, $dy/dx=-F_x/F_y$. Therefore, the verified correct option is $-F_x/F_y$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
Logarithmic differentiation: if $y=\frac{x^2\sqrt{x+1}}{e^x}$, then $y'=$

a. $y\left(\frac{2}{x}+\frac{1}{2(x+1)}-1\right)$ ✓

b. $y\left(\frac{2}{x}-\frac{1}{2(x+1)}-1\right)$

c. $y\left(\frac{2}{x}+\frac{1}{2(x+1)}+1\right)$

d. $y\left(2x+\frac{1}{2}-1\right)$

$\ln y=2\ln x+\frac{1}{2}\ln(x+1)-x$. Differentiate: $y'/y=2/x+1/(2(x+1))-1$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
Logarithmic differentiation: $y=x^x$. $y'=$

a. $x^x(1+\ln x)$ ✓

b. $x\cdot x^{x-1}$

c. $x^x\ln x$

d. e^x

$\ln y=x\ln x\Rightarrow y'/y=1+\ln x\Rightarrow y'=x^x(1+\ln x)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Differentiation of geometric series $1/(1-x) = \sum x^n$ gives $1/(1-x)^2 =$

a. $\sum n x^{n-1}$

b. $\sum (n+1)x^n$ ✓

c. $\sum x^{n-1}$

d. $\sum n x^n$

$d/dx \sum x^n = \sum n x^{n-1} = \sum (n+1)x^n$ (shift n).

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Leibniz rule: $(fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n-k)}$. Analogy with:

a. **binomial theorem** ✓

b. FTC

c. MVT

d. Taylor series

Leibniz general product rule ↔ binomial theorem.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Term-by-term differentiation of $\sum a_n x^n$ within $|x| < R$ gives:

a. $\sum n a_n x^n$

b. $\sum a_{n+1} x^{n+1}$

c. $\sum a_n x^{n+1}/(n+1)$

d. $\sum n a_n x^{n-1}$ ✓

Differentiate term-by-term: $\sum n a_n x^{n-1}$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

L'Hôpital's rule applies to $\lim_{x \to a} \frac{f}{g}$ when the limit is of the form:

a. 0/1

b. 1/0

c. 0/0 ✓

d. 1/1

Indeterminate form $0/0$ or ∞/∞.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A plane region R is bounded by the graph of $y = e^{-x^2}$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the line $y = -2$. The cross-sections of the solid perpendicular to the x-axis are squares. The volume of this solid is given by $V = \int_a^b A(x)dx$. (a) Find the outer radius and inner radius of a representative disk if the region were revolved about the line $y = -2$ using the washer method. (b) Set up, but do not evaluate, an integral expression for the volume of the solid generated by revolving the region R about the line $y = -2$ using the washer method. (c) Set up, but do not evaluate, an integral expression for the volume of the solid generated by taking cross-sections perpendicular to the x-axis that are squares. (d) Explain why the integral expression in part (c) is the correct representation for the volume of the solid.

a. The radius of a disk in the washer method is $R(x) = e^{-x^2} + 2$ and $r(x) = 2$.

b. The volume using the washer method is $V = \int_0^1 \pi((e^{-x^2} + 2)^2 - 2^2)dx$.

c. The area of a square cross-section is $A(x) = (e^{-x^2})^2$.

d. The integral expression in part (c) represents the volume because it sums the areas of square cross-sections perpendicular to the x-axis over the interval [0, 1]. ✓

(a) When revolving the region R about the line $y = -2$, the outer radius $R(x)$ is the distance from $y = -2$ to the upper boundary of the region $y = e^{-x^2}$, which is $R(x) = e^{-x^2} - (-2) = e^{-x^2} + 2$. The inner radius $r(x)$ is the distance from $y = -2$ to the lower boundary of the region (the x-axis, $y = 0$), which is $r(x) = 0 - (-2) = 2$. (b) Using the washer method, the volume V is given by the integral of the area of the washers, which is $\pi(R(x)^2 - r(x)^2)$. Therefore, the integral express.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider a region R in the xy-plane bounded by the graphs of $y = \frac{1}{2}x^2$ and $y = 2x$. A solid has its base in region R, and for each value of x in the interval $[0, 4]$, the cross-section perpendicular to the x-axis is an equilateral triangle with side length $s(x) = 2x - \frac{1}{2}x^2$. Write an integral expression that represents the volume of this solid. You do not need to evaluate the integral.

a. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right)^2 dx$ ✓

b. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right) dx$

c. $V = \int_0^4 \frac{1}{2}\left(2x - \frac{1}{2}x^2\right)^2 dx$

d. $V = \int_0^4 \frac{\sqrt{3}}{2}\left(2x - \frac{1}{2}x^2\right) dx$

The base of the solid is defined by the region R bounded by $y = \frac{1}{2}x^2$ and $y = 2x$. The upper bound is $y = 2x$ and the lower bound is $y = \frac{1}{2}x^2$. The intersection points are found by setting $\frac{1}{2}x^2 = 2x$, which gives $x^2 = 4x$, so $x^2 - 4x = 0$, and $x(x - 4) = 0$. Thus, the region extends from $x = 0$ to $x = 4$. The side length of the equilateral triangle cross-section is given by $s(x) = 2x - \frac{1}{2}x^2$. The area of an equilateral triangle with side length s is.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x)$ whose graph is shown below. The graph consists of a line segment and a quarter circle. The line segment connects the points $(2, 4)$ and $(4, 0)$. The quarter circle is in the second quadrant and is centered at $(0, 0)$ with radius 2. Let $A(x)$ be the area of the region bounded by the graph of $f(t)$, the t-axis, and the lines $t = 0$ and $t = x$. (a) Find $A(3)$.

a. **4** ✓

b. 6

c. 5

d. 7

The function $f(x)$ is defined piecewise. For $0 \le t \le 2$, $f(t)$ is the upper part of a circle centered at the origin with radius 2, so $f(t) = \sqrt{4 - t^2}$. For $2 < t \le 4$, $f(t)$ is a line segment connecting $(2, 4)$ and $(4, 0)$. The equation of this line can be found using the two-point form: $y - 0 = \frac{4-0}{2-4}(x - 4)$, which simplifies to $y = -2(x - 4)$, so $f(t) = -2(t - 4)$. To find $A(3)$, we need to calculate the definite integral $\int_0^3 f(t)\,dt$. We split this integral i.

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36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The graph of a differentiable function f is shown in the figure above. The function f has a local maximum at $x = c$ and a local minimum at $x = e$. The point $(c, f(c))$ is approximately $(1, 4)$ and the point $(e, f(e))$ is approximately $(4, -2)$. The shaded region, bounded by the graph of f, the x-axis, and the lines $x = 0$ and $x = 5$, has area 12. $*If*\int_{0}^{c} f(x) \, dx = k, *whatisthevalueof*\int_{c}^{5} f(x) \, dx*intermsof*k$?

a. 12 + k

b. **12 - k ✓**

c. k - 12

d. k + 2

The total area of the shaded region from $x = 0$ to $x = 5$ is given as 12. This total area can be expressed as the sum of the areas of two sub-regions: the area from $x = 0$ to $x = c$ and the area from $x = c$ to $x = 5$. Mathematically, this can be written using definite integrals: $\int_0^5 f(x)\,dx = \int_0^c f(x)\,dx + \int_c^5 f(x)\,dx$. We are given that the total area is 12, so $\int_0^5 f(x)\,dx = 12$. We are also given that $\int_0^c f(x)\,dx = k$. Substituting these values.

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = a$ and a local minimum at $x = b$. The tangent line to the graph of f at $x = c$ has the equation $y - f(c) = \frac{1}{2}(x - c)$. Given that $f(c) = 3$ and $f'(c) = \frac{1}{2}$, find the value of c.

a. 4

b. 3

c. **2 ✓**

d. 1

The equation of the tangent line to the graph of a function f at $x = c$ is given by $y - f(c) = f'(c)(x - c)$. We are given that the equation of the tangent line at $x = c$ is $y - f(c) = \frac{1}{2}(x - c)$. Comparing this to the general form, we see that $f'(c) = \frac{1}{2}$. We are also given that $f(c) = 3$. The graph shows a point on the function $f(x)$ at $(x, y) = (2, 3)$. Since the point $(c, f(c))$ must lie on the graph of f, we have $c = 2$ and $f(c) = 3$. Therefore, the value of c is 2.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Consider the function $f(x) = x^2 - 6x + 10$. Find the x-coordinates of the points on the graph of $y = f(x)$ where the tangent line is horizontal.

a. x = 3 ✓

b. x = -3 and x = 3

c. x = 1 and x = 5

d. x = 1

A horizontal tangent line occurs where the derivative of the function is equal to zero. First, find the derivative of $f(x)$: $f'(x) = \frac{d}{dx}(x^2 - 6x + 10) = 2x - 6$. Next, set the derivative equal to zero and solve for x : $2x - 6 = 0$. Adding 6 to both sides gives $2x = 6$. Dividing by 2 gives $x = 3$. Therefore, the x-coordinate of the point where the tangent line is horizontal is $x = 3$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 15.01 Double Integrals

A differentiable function f is defined on the interval $[-4, 4]$. The graph of f', the derivative of f, is shown in the figure above. The graph of f' consists of two line segments and a semicircle. (a) Find the value of $f(2) - f(-2)$. Show the setup for your answer. (b) For how many values of c in the interval $(-4, 4)$ does $f'(c) = 0$? Explain your reasoning. (c) The function f has a local maximum at $x = k$. Find the value of k. Explain your reasoning. (d) The function f has an inflection point at $x = p$. Find the value of p. Explain your reasoning.

a. There are no values of c for which $f'(c) = 0$.

b. There is one value of c for which $f'(c) = 0$.

c. There are two values of c for which $f'(c) = 0$. ✓

d. There are three values of c for which $f'(c) = 0$.

(a) By the Fundamental Theorem of Calculus, $f(2) - f(-2) = \int_{-2}^{2} f'(x)\,dx$. The region from $x = -2$ to $x = 2$ is a semicircle with radius 2. The area of this semicircle is $\frac{1}{2}\pi(2)^2 = 2\pi$. Therefore, $f(2) - f(-2) = 2\pi$. **(b)** The graph of $f'(x)$ is shown. $f'(c) = 0$ when the graph of f' intersects the x-axis. This occurs at $x = -4$ and $x = 2$. Since we are looking for values of c in the open interval $(-4, 4)$, there is one such value, $c = 2$. **(c)** A local maximum o.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 16.01 Vector Fields and Line Integrals

A particle moves along the x-axis. Its velocity is given by the function $v(t) = \sqrt{t^2 + 1}$ for $t \geq 0$. The position of the particle at time $t = 0$ is $x(0) = 5$. Consider the function $P(x)$ defined as the total displacement of the particle from $t = 0$ to $t = x$. That is, $P(x) = x(x) - x(0)$. Which of the following statements about $P(x)$ is true for $x > 0$?

a. $P(x)$ **is always positive and increasing.** ✓

b. $P(x)$ is always negative and decreasing.

c. $P(x)$ is always positive and decreasing.

d. $P(x)$ is always negative and increasing.

The velocity of the particle is given by $v(t) = \sqrt{t^2 + 1}$. Since $t \geq 0$, $t^2 \geq 0$, so $t^2 + 1 \geq 1$. Therefore, $v(t) = \sqrt{t^2 + 1} \geq \sqrt{1} = 1$ for all $t \geq 0$. This means the velocity is always positive. The position of the particle is given by $x(t)$. The Fundamental Theorem of Calculus states that the change in position (displacement) is the integral of the velocity function: $x(t) - x(0) = \int_0^t v(u)\,du$. The function $P(x)$ represents the total displacement from $t = 0$ to.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Evaluating Limits

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 8

d. 16

The area of the region bounded by two curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$, where $f(x) \geq g(x)$ on $[a, b]$, is given by the integral $\int_a^b (f(x) - g(x))dx$. First, we find the intersection points of $y = x^2$ and $y = 4$ by setting them equal: $x^2 = 4$, which gives $x = -2$ and $x = 2$. So, our limits of integration are $a = -2$ and $b = 2$. On the interval $[-2, 2]$, the line $y = 4$ is above the parabola $y = x^2$. Therefore, the area A is given by: $A = \int_{-2}^{2} (4 - x^2)dx$ Now, we eva.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.02 Continuity

The graph of the function $f(x)$ is shown below. What is the value of the definite integral $\int_{-2}^{2} f(x)\,dx$?

a. 0

b. 2

c. **4** ✓

d. 8

The definite integral $\int_{-2}^{2} f(x)\,dx$ represents the net signed area between the graph of $f(x)$ and the x-axis from $x = -2$ to $x = 2$. The graph of $f(x)$ for $-2 \le x \le 2$ forms a semicircle above the x-axis with radius 2. The area of this semicircle is $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(2^2) = 2\pi$. However, the provided figure shows a region that is a combination of triangles and a rectangle. From $x = -2$ to $x = 0$, the area is a triangle with base 2 and height 2, so the area is.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 The Derivative from First Principles

A particle moves along a straight line so that its velocity at time $t \ge 0$ is modeled by $v(t) = 6t^2 - 18t + 12$. At what time t is the acceleration of the particle equal to 0?

a. $t = \frac{3}{2}$ ✓

b. $t = 1$

c. $t = 2$

d. $t = 3$

To find the acceleration $a(t)$, take the derivative of the velocity function $v(t)$ with respect to time t: $a(t) = v'(t) = \frac{d}{dt}(6t^2 - 18t + 12) = 12t - 18$. Set $a(t) = 0$ and solve for t: $12t - 18 = 0 \Rightarrow 12t = 18 \Rightarrow t = \frac{18}{12} = \frac{3}{2}$. Common Misconceptions: - Solving $v(t) = 0$ yields $6(t-1)(t-2) = 0 \Rightarrow t = 1$ or $t = 2$. These are the times when the particle is at rest, not when its acceleration is zero. - Omitting the factor of 2 when applying the po.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Consider the function $f(x)$ whose derivative is given by $f'(x) = \frac{\sin(x)}{x^2+1}$. A related function $g(x)$ is defined by $g(x) = \int_0^x f(t)dt$. If $f(0) = k$, where k is a positive constant, find the value of $g'(0)$ and determine if $g(x)$ is increasing or decreasing at $x = 0$.

a. $g'(0) = k$**, and** $g(x)$ **is increasing.** ✓

b. $g'(0) = 0$, and $g(x)$ is neither increasing nor decreasing.

c. $g'(0) = 1$, and $g(x)$ is increasing.

d. $g'(0) = k$, and $g(x)$ is decreasing.

We are given $g(x) = \int_0^x f(t)dt$. By the Fundamental Theorem of Calculus Part 1, the derivative of $g(x)$ is $g'(x) = f(x)$. We need to find $g'(0)$. Substituting $x = 0$ into $g'(x)$, we get $g'(0) = f(0)$. We are given that $f(0) = k$, where k is a positive constant. Therefore, $g'(0) = k$. To determine if $g(x)$ is increasing or decreasing at $x = 0$, we examine the sign of $g'(0)$. Since k is a positive constant, $g'(0) = k > 0$. A function is increasing at a point if its derivative at that point is posit.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Consider the function $f(x)$ whose graph is shown above. The graph of $f(x)$ consists of line segments and a quarter circle. Let $g(x) = \int_0^x f(t)\,dt$.

a. **g'(2) = 2** ✓

b. g'(2) = 0

c. g'(5) = -2

d. g'(5) = 0

By the Fundamental Theorem of Calculus Part 1, $g'(x) = f(x)$. We need to find $g'(2)$, which means we need to evaluate $f(2)$. From the graph, at $x = 2$, the function $f(x)$ is on a line segment connecting $(0, 4)$ and $(4, 0)$. The equation of this line segment can be found. The slope is $m = \frac{0-4}{4-0} = -1$. Using the point-slope form with $(4, 0)$, we get $y - 0 = -1(x - 4)$, which simplifies to $y = -x + 4$. Therefore, $f(x) = -x + 4$ for $0 \le x \le 4$. Evaluating $f(2)$ gives $f(2) = -2 + 4 = 2$..

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